

CONTENT MONETIZATION MODELER

ABSTRACT

The Content Monetization Modeler is a machine learning project developed using Python, Pandas, Scikit-learn, and Streamlit to predict YouTube advertisement revenue based on video and channel characteristics such as views, watch time, subscribers, category, device, and country. The project involves data preprocessing, exploratory data analysis, feature engineering, and Linear Regression model development to generate accurate revenue predictions. The trained model is deployed through an interactive Streamlit application, enabling users to estimate potential earnings instantly and helping content creators make informed, data-driven decisions to optimize their content strategy and maximize monetization opportunities.

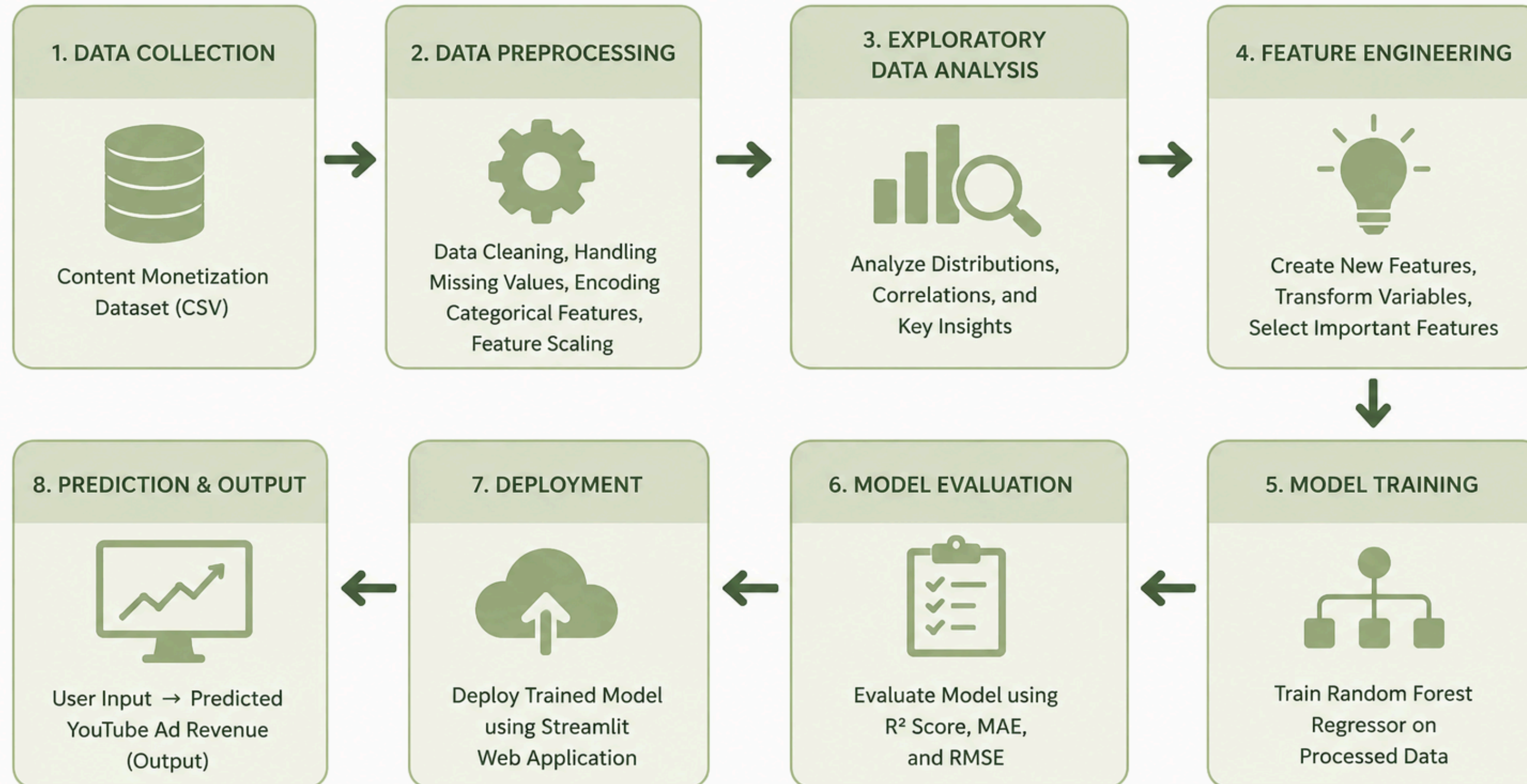
PROBLEM STATEMENT

1. Predicting YouTube advertisement revenue is challenging because it depends on multiple factors such as views, watch time, subscribers, content category, device type, and audience location.
2. A machine learning-based system is needed to accurately estimate potential ad revenue, helping content creators make data-driven decisions and optimize their monetization strategy.

DATASET OVERVIEW

Property	Details
Dataset Name	<u>YouTube Monetization Modeler</u>
Source	[Synthetic, created for learning purposes]
Format	CSV (Comma-Separated Values)
Total Records	<i>~122,000 rows</i>
Key Features	Views, Watch Time, Subscribers, Category, Device, Country
Target Variable	Estimated Ad Revenue
Dataset Status	Cleaned and Preprocessed for Machine Learning

WORKFLOW



KEY ANALYTICAL STEPS

1. Performed data cleaning and handled missing values to prepare the dataset for analysis.
2. Conducted Exploratory Data Analysis (EDA) to identify patterns, trends, and relationships between features.
3. Applied feature engineering, label encoding, and data scaling to improve model performance.
4. Trained and evaluated a Linear Regression model using R^2 Score, MAE, and RMSE to predict YouTube advertisement revenue accurately.

MODEL PERFORMANCE

Algorithm Used : Random Forest Regressor

Metric	Value
R² Score	0.9485
Mean Absolute Error (MAE)	3.67
Root Mean Squared Error (RMSE)	14.05

STREAMLIT DASHBOARD

Deploy

 Project Information

Model Used: Random Forest Regressor

R² Score: 0.9485

MAE: 3.67

RMSE: 14.05

YouTube Ad Revenue Predictor 

Predict YouTube Ad Revenue using Machine Learning

Views

0

-

+

Likes

0

-

+

Comments

0

-

+

Watch Time Minutes

0.00

-

+

Video Length Minutes

-

+

CONCLUSION

- The Content Monetization Modeler successfully predicts YouTube advertisement revenue using a Random Forest Regressor with high accuracy (R^2 Score: 0.9485). By combining data preprocessing, feature engineering, machine learning, and Streamlit deployment, the project provides an efficient and user-friendly solution for estimating potential earnings, enabling content creators to make informed, data-driven decisions and optimize their monetization strategies.

THANK
YOU